

Parametric Equations

ANSWER KEY



Derivatives of parametric equations

- 1 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t .

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2}.$$

- 2 For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.

$$\frac{dy}{dx} = \frac{\cos t}{-\sin t} = -\cot t. \text{ At } t = \frac{\pi}{4}, -\cot\left(\frac{\pi}{4}\right) = -1.$$

Second derivatives of parametric equations

- 3 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{d^2y}{dx^2}$ using $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$.

$$\frac{dy}{dx} = \frac{3t}{2}, \frac{d}{dt}\left(\frac{3t}{2}\right) = \frac{3}{2}, \frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t}.$$

Tangent lines to parametric curves

- 4 Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.

$$\text{At } t = 2: x = 3, y = 6. \frac{dx}{dt} = 2t = 4; \frac{dy}{dt} = 3t^2 - 1 = 11. \frac{dy}{dx} = \frac{11}{4}. \text{ Tangent: } y - 6 = \frac{11}{4}(x - 3).$$

Arc length of parametric curves

- 5 State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

- 6 Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.

$$\frac{dx}{dt} = 3, \frac{dy}{dt} = 4. L = \int_0^2 \sqrt{9 + 16} dt = \int_0^2 5 dt = 10.$$

- 7 Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.

$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 3t^2. L = \int_0^1 \sqrt{(2t)^2 + (3t^2)^2} dt = \int_0^1 \sqrt{4t^2 + 9t^4} dt.$$

Speed as a vector quantity

- 8 A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.

$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 2. \text{ At } t = 1: \frac{dx}{dt} = 2, \frac{dy}{dt} = 2. \text{ Speed} = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}.$$

- 9 A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t , and interpret the result.

$$\frac{dx}{dt} = -3\sin t, \frac{dy}{dt} = 3\cos t. \text{ Speed} = \sqrt{9\sin^2 t + 9\cos^2 t} = \sqrt{9} = 3 \text{ (constant). The particle traces a circle of radius 3 at constant speed.}$$

Eliminating the parameter

- 10 Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.

Since $x = t$, substitute directly: $y = x^2 - 1$, an upward parabola shifted down 1 unit.

- 11 Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.

From $x = 2t + 1$: $t = \frac{x-1}{2}$. Substitute: $y = 4\left(\frac{x-1}{2}\right) - 3 = 2(x-1) - 3 = 2x - 5$. A line with slope 2.

- 12 Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.

Substituting gives $x^2 + y^2 = 1$, the equation of a unit circle. For most x -values in $(-1, 1)$, there are two corresponding y -values ($\pm\sqrt{1-x^2}$), so this relation fails the vertical line test and is not a function of x .

Concavity of parametric curves

- 13 For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.

For $t > 0$: $\frac{d^2y}{dx^2} = \frac{3}{4t} > 0$, so the curve is concave up. For $t < 0$: $\frac{3}{4t} < 0$, so the curve is concave down.

Horizontal and vertical tangents

- 14 For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).

$\frac{dy}{dt} = 3t^2 - 3 = 0 \Rightarrow t = \pm 1$. Check $\frac{dx}{dt} = 2t$: at $t = \pm 1$, $\frac{dx}{dt} = \pm 2 \neq 0$, so both give valid horizontal tangents.

- 15 For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).

$\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. Check $\frac{dy}{dt} = 3(0)^2 - 3 = -3 \neq 0$, so $t = 0$ gives a valid vertical tangent, at the point $x(0) = -4, y(0) = 0$.

Motion along a parametric path (particle interpretation)

- 16 A particle's position is $x(t) = t^2, y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).

$\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. At $t = 0$: position is $(0, 0)$, and the particle is momentarily moving purely vertically (since horizontal velocity is 0 there).

- 17 For the particle above, find the speed at $t = 2$.

$\frac{dx}{dt} = 2t = 4; \frac{dy}{dt} = 3t^2 - 3 = 9$. Speed $= \sqrt{4^2 + 9^2} = \sqrt{16 + 81} = \sqrt{97} \approx 9.85$.

Arc length of a full circle (verifying the circumference formula)

- 18 Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r \cos t, y(t) = r \sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.

$\frac{dx}{dt} = -r \sin t, \frac{dy}{dt} = r \cos t$. $L = \int_0^{2\pi} \sqrt{r^2 \sin^2 t + r^2 \cos^2 t} dt = \int_0^{2\pi} r dt = r(2\pi) = 2\pi r$, matching the known circumference formula.

- 19 Find the arc length of the parametric curve $x(t) = e^t \cos t, y(t) = e^t \sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.

$\frac{dx}{dt} = e^t \cos t - e^t \sin t, \frac{dy}{dt} = e^t \sin t + e^t \cos t$. After simplification, $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 2e^{2t}$. So $L = \int_0^\pi \sqrt{2e^{2t}} dt = \int_0^\pi e^t \sqrt{2} dt$.

Parametric equations from real motion

- 20 A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.

Angular speed $= \frac{2\pi}{40} = \frac{\pi}{20}$ rad/s. Starting at the bottom means height $= 12 - 10 \cos\left(\frac{\pi}{20}t\right)$ (using $x(t) = t$ for time, $y(t) = 12 - 10 \cos\left(\frac{\pi}{20}t\right)$ for height).

- 21 For the Ferris wheel model above, find the rider's vertical velocity $\frac{dy}{dt}$ at $t = 10$ s (a quarter revolution).

$\frac{dy}{dt} = 10 \cdot \frac{\pi}{20} \sin\left(\frac{\pi}{20}t\right) = \frac{\pi}{2} \sin\left(\frac{\pi}{20}t\right)$. At $t = 10$: $\frac{\pi}{2} \sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \approx 1.57$ m/s.

Multiple choice — easy 15

22 For $x(t) = 6t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $-\frac{3}{2}$ ✓

b. -6

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-6t^2}{12t} = \frac{-6}{12}t. \text{ At } t = 3: \frac{-6}{12}(3) = -\frac{3}{2}.$$

23 For $x(t) = 5t^2$, $y(t) = -5t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. $-\frac{3}{2}$ ✓

b. $-\frac{1}{2}$

c. $-\frac{2}{3}$

d. $\frac{3}{2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-15t^2}{10t} = \frac{-15}{10}t. \text{ At } t = 1: \frac{-15}{10}(1) = -\frac{3}{2}.$$

24 For $x(t) = 2t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. -4

b. 9

c. -9 ✓

d. -8

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-6t^2}{4t} = \frac{-6}{4}t. \text{ At } t = 6: \frac{-6}{4}(6) = -9.$$

25 For $x(t) = 6t^2$, $y(t) = 2t^3$, find $\frac{dy}{dx}$ at $t = 2$.

a. 1 ✓

b. 2

c. -1

d. 4

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{6t^2}{12t} = \frac{6}{12}t. \text{ At } t = 2: \frac{6}{12}(2) = 1.$$

26 For $x(t) = 4t^2$, $y(t) = -2t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{9}{4}$

b. -4

c. $-\frac{9}{4}$ ✓

d. $-\frac{5}{4}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-6t^2}{8t} = \frac{-6}{8}t. \text{ At } t = 3: \frac{-6}{8}(3) = -\frac{9}{4}.$$

27 For $x(t) = t^2$, $y(t) = -6t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. -8

b. -9 ✓

c. 9

d. $-\frac{1}{9}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-18t^2}{2t} = \frac{-18}{2}t. \text{ At } t = 1: \frac{-18}{2}(1) = -9.$$

28 For $x(t) = 4t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{5}{2}$

b. $\frac{3}{2}$ ✓

c. $\frac{32}{3}$

d. $-\frac{3}{2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{8t} = \frac{3}{8}t. \text{ At } t = 4: \frac{3}{8}(4) = \frac{3}{2}.$$

29 For $x(t) = 5t^2$, $y(t) = 6t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. $-\frac{54}{5}$

b. $\frac{10}{3}$

c. $\frac{59}{5}$

d. $\frac{54}{5}$ ✓

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{18t^2}{10t} = \frac{18}{10}t. \text{ At } t = 6: \frac{18}{10}(6) = \frac{54}{5}.$$

30 For $x(t) = 3t^2$, $y(t) = -t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{3}{2}$

b. $-\frac{1}{2}$

c. -6

d. $-\frac{3}{2}$ ✓

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-3t^2}{6t} = \frac{-3}{6}t. \text{ At } t = 3: \frac{-3}{6}(3) = -\frac{3}{2}.$$

31 For $x(t) = 6t^2$, $y(t) = -3t^3$, find $\frac{dy}{dx}$ at $t = 1$.

a. $-\frac{4}{3}$

b. $-\frac{3}{4}$ ✓

c. $\frac{1}{4}$

d. $\frac{3}{4}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-9t^2}{12t} = \frac{-9}{12}t. \text{ At } t = 1: \frac{-9}{12}(1) = -\frac{3}{4}.$$

32 For $x(t) = 6t^2$, $y(t) = 4t^3$, find $\frac{dy}{dx}$ at $t = 6$.

a. -6

b. 6 ✓

c. 7

d. $\frac{13}{2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{12t^2}{12t} = \frac{12}{12}t. \text{ At } t = 6: \frac{12}{12}(6) = 6.$$

33 For $x(t) = t^2$, $y(t) = 5t^3$, find $\frac{dy}{dx}$ at $t = 4$.

a. $\frac{8}{15}$

b. -30

c. 31

d. 30 ✓

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{15t^2}{2t} = \frac{15}{2}t. \text{ At } t = 4: \frac{15}{2}(4) = 30.$$

34 For $x(t) = 3t^2$, $y(t) = -6t^3$, find $\frac{dy}{dx}$ at $t = 3$.

a. 9

b. -8

c. -9 ✓

d. -1

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-18t^2}{6t} = -\frac{18}{6}t. \text{ At } t = 3: -\frac{18}{6}(3) = -9.$$

35 For $x(t) = 6t^2$, $y(t) = 2t^3$, find $\frac{dy}{dx}$ at $t = 5$.

a. $-\frac{5}{2}$

b. $\frac{7}{2}$

c. 10

d. $\frac{5}{2}$ ✓

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{6t^2}{12t} = \frac{6}{12}t. \text{ At } t = 5: \frac{6}{12}(5) = \frac{5}{2}.$$

36 For $x(t) = 5t^2$, $y(t) = -4t^3$, find $\frac{dy}{dx}$ at $t = 3$.

- a. $-\frac{13}{5}$
- b. $\frac{18}{5}$
- c. $-\frac{18}{5}$ ✓
- d. $-\frac{5}{2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-12t^2}{10t} = \frac{-12}{10}t. \text{ At } t = 3: \frac{-12}{10}(3) = -\frac{18}{5}.$$

Multiple choice — medium 15

37 Eliminate the parameter from $x(t) = -2t - 3$, $y(t) = 7t - 9$ to find the Cartesian equation, then state its slope.

- a. $-\frac{7}{2}$ ✓
- b. $-\frac{5}{2}$
- c. $-\frac{2}{7}$
- d. $\frac{7}{2}$

From $x = -2t - 3$: $t = \frac{x+3}{-2}$. Substituting into $y = 7t - 9$ gives a line of slope $\frac{7}{-2} = -\frac{7}{2}$.

38 Eliminate the parameter from $x(t) = 6t + 6$, $y(t) = -9t + 8$ to find the Cartesian equation, then state its slope.

- a. $-\frac{3}{2}$ ✓
- b. $-\frac{2}{3}$
- c. $-\frac{1}{2}$
- d. $\frac{3}{2}$

From $x = 6t + 6$: $t = \frac{x-6}{6}$. Substituting into $y = -9t + 8$ gives a line of slope $\frac{-9}{6} = -\frac{3}{2}$.

- 39 Eliminate the parameter from $x(t) = 4t - 1$, $y(t) = 7t + 6$ to find the Cartesian equation, then state its slope.

a. $-\frac{7}{4}$

b. $\frac{11}{4}$

c. $\frac{7}{4}$ ✓

d. $\frac{4}{7}$

From $x = 4t - 1$: $t = \frac{x+1}{4}$. Substituting into $y = 7t + 6$ gives a line of slope $\frac{7}{4} = \frac{7}{4}$.

- 40 Eliminate the parameter from $x(t) = 2t - 9$, $y(t) = -t + 4$ to find the Cartesian equation, then state its slope.

a. $-\frac{1}{2}$ ✓

b. -2

c. $\frac{1}{2}$

d. 0

From $x = 2t - 9$: $t = \frac{x+9}{2}$. Substituting into $y = -t + 4$ gives a line of slope $\frac{-1}{2} = -\frac{1}{2}$.

- 41 Eliminate the parameter from $x(t) = t - 6$, $y(t) = -9t + 3$ to find the Cartesian equation, then state its slope.

a. $-\frac{1}{9}$

b. 9

c. -8

d. -9 ✓

From $x = t - 6$: $t = \frac{x+6}{1}$. Substituting into $y = -9t + 3$ gives a line of slope $\frac{-9}{1} = -9$.

- 42 Eliminate the parameter from $x(t) = -5t - 4$, $y(t) = -4t - 7$ to find the Cartesian equation, then state its slope.

a. $-\frac{4}{5}$

b. $\frac{5}{4}$

c. $\frac{4}{5}$ ✓

d. $\frac{9}{5}$

From $x = -5t - 4$: $t = \frac{x+4}{-5}$. Substituting into $y = -4t - 7$ gives a line of slope $\frac{-4}{-5} = \frac{4}{5}$.

- 43 Eliminate the parameter from $x(t) = 4t - 7$, $y(t) = -5t + 9$ to find the Cartesian equation, then state its slope.

a. $-\frac{4}{5}$

b. $-\frac{5}{4}$ ✓

c. $-\frac{1}{4}$

d. $\frac{5}{4}$

From $x = 4t - 7$: $t = \frac{x+7}{4}$. Substituting into $y = -5t + 9$ gives a line of slope $\frac{-5}{4} = -\frac{5}{4}$.

- 44 Eliminate the parameter from $x(t) = t + 1$, $y(t) = 2t + 5$ to find the Cartesian equation, then state its slope.

a. 3

b. -2

c. $\frac{1}{2}$

d. 2 ✓

From $x = t + 1$: $t = \frac{x-1}{1}$. Substituting into $y = 2t + 5$ gives a line of slope $\frac{2}{1} = 2$.

- 45 Eliminate the parameter from $x(t) = 6t - 5$, $y(t) = -2t + 5$ to find the Cartesian equation, then state its slope.

a. $\frac{2}{3}$

b. $-\frac{1}{3}$ ✓

c. -3

d. $\frac{1}{3}$

From $x = 6t - 5$: $t = \frac{x+5}{6}$. Substituting into $y = -2t + 5$ gives a line of slope $\frac{-2}{6} = -\frac{1}{3}$.

- 46 Eliminate the parameter from $x(t) = 5t - 6$, $y(t) = -3t + 5$ to find the Cartesian equation, then state its slope.

a. $-\frac{5}{3}$

b. $\frac{3}{5}$

c. $-\frac{3}{5}$ ✓

d. $\frac{2}{5}$

From $x = 5t - 6$: $t = \frac{x+6}{5}$. Substituting into $y = -3t + 5$ gives a line of slope $\frac{-3}{5} = -\frac{3}{5}$.

- 47 Eliminate the parameter from $x(t) = -4t + 3$, $y(t) = -t + 5$ to find the Cartesian equation, then state its slope.

a. $-\frac{1}{4}$

b. $\frac{1}{4}$ ✓

c. 4

d. $\frac{5}{4}$

From $x = -4t + 3$: $t = \frac{x-3}{-4}$. Substituting into $y = -t + 5$ gives a line of slope $\frac{-1}{-4} = \frac{1}{4}$.

- 48 Eliminate the parameter from $x(t) = t + 6$, $y(t) = -7t - 4$ to find the Cartesian equation, then state its slope.

a. -7 ✓

b. $-\frac{1}{7}$

c. 7

d. -6

From $x = t + 6$: $t = \frac{x-6}{1}$. Substituting into $y = -7t - 4$ gives a line of slope $\frac{-7}{1} = -7$.

- 49 Eliminate the parameter from $x(t) = 4t + 1$, $y(t) = 9t + 7$ to find the Cartesian equation, then state its slope.

a. $-\frac{9}{4}$

b. $\frac{13}{4}$

c. $\frac{9}{4}$ ✓

d. $\frac{4}{9}$

From $x = 4t + 1$: $t = \frac{x-1}{4}$. Substituting into $y = 9t + 7$ gives a line of slope $\frac{9}{4} = \frac{9}{4}$.

- 50 Eliminate the parameter from $x(t) = -6t$, $y(t) = -6t + 9$ to find the Cartesian equation, then state its slope.

a. 1 ✓

b. 2

c. $\frac{3}{2}$

d. -1

From $x = -6t$: $t = \frac{x}{-6}$. Substituting into $y = -6t + 9$ gives a line of slope $\frac{-6}{-6} = 1$.

- 51 Eliminate the parameter from $x(t) = 3t - 1$, $y(t) = 2t + 2$ to find the Cartesian equation, then state its slope.

- a. $-\frac{2}{3}$
- b. $\frac{2}{3}$ ✓
- c. $\frac{5}{3}$
- d. $\frac{3}{2}$

From $x = 3t - 1$: $t = \frac{x+1}{3}$. Substituting into $y = 2t + 2$ gives a line of slope $\frac{2}{3} = \frac{2}{3}$.

Multiple choice — hard 15

- 52 For $x(t) = 3t^2$, $y(t) = 2t^3 + 6t$, find $\frac{dy}{dx}$ at $t = 4$.

- a. $\frac{21}{4}$
- b. $\frac{17}{4}$ ✓
- c. $\frac{4}{17}$
- d. $-\frac{17}{4}$

$\frac{dx}{dt} = 6t$, $\frac{dy}{dt} = 6t^2 + 6$. At $t = 4$: $\frac{dx}{dt} = 24$, $\frac{dy}{dt} = 102$. $\frac{dy}{dx} = \frac{102}{24} = \frac{17}{4}$.

- 53 For $x(t) = 3t^2 + 5$, $y(t) = t^3 - 4t$, find $\frac{dy}{dx}$ at $t = 1$.

- a. $\frac{1}{6}$
- b. $\frac{5}{6}$
- c. -6
- d. $-\frac{1}{6}$ ✓

$\frac{dx}{dt} = 6t$, $\frac{dy}{dt} = 3t^2 - 4$. At $t = 1$: $\frac{dx}{dt} = 6$, $\frac{dy}{dt} = -1$. $\frac{dy}{dx} = \frac{-1}{6} = -\frac{1}{6}$.

54 For $x(t) = 5t^2 + 4$, $y(t) = 2t^3 + 5t$, find $\frac{dy}{dx}$ at $t = 4$.

a. $-\frac{101}{40}$

b. $\frac{141}{40}$

c. $\frac{40}{101}$

d. $\frac{101}{40}$ ✓

$\frac{dx}{dt} = 10t$, $\frac{dy}{dt} = 6t^2 + 5$. At $t = 4$: $\frac{dx}{dt} = 40$, $\frac{dy}{dt} = 101$. $\frac{dy}{dx} = \frac{101}{40} = \frac{101}{40}$.

55 For $x(t) = 2t^2 - 6$, $y(t) = 5t^3 + 3t$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{25}{2}$

b. $\frac{23}{2}$ ✓

c. $\frac{2}{23}$

d. $-\frac{23}{2}$

$\frac{dx}{dt} = 4t$, $\frac{dy}{dt} = 15t^2 + 3$. At $t = 3$: $\frac{dx}{dt} = 12$, $\frac{dy}{dt} = 138$. $\frac{dy}{dx} = \frac{138}{12} = \frac{23}{2}$.

56 For $x(t) = 3t^2 - 5$, $y(t) = 5t^3 + 3t$, find $\frac{dy}{dx}$ at $t = 3$.

a. $\frac{26}{3}$

b. $\frac{3}{23}$

c. $-\frac{23}{3}$

d. $\frac{23}{3}$ ✓

$\frac{dx}{dt} = 6t$, $\frac{dy}{dt} = 15t^2 + 3$. At $t = 3$: $\frac{dx}{dt} = 18$, $\frac{dy}{dt} = 138$. $\frac{dy}{dx} = \frac{138}{18} = \frac{23}{3}$.

57 For $x(t) = t^2 + 2t$, $y(t) = t^3 + 6t^2$, find the larger value of t where the tangent line is horizontal.

a. -4

b. 0 ✓

c. -3

d. 3

$\frac{dy}{dt} = 3t^2 + 12t = 3(t - (-4))(t - (0)) = 0$ gives $t = -4$ and $t = 0$; the larger is 0 .

58 For $x(t) = t^2$, $y(t) = t^3 - 9t^2 + 24t$, find the larger value of t where the tangent line is horizontal.

a. 2

b. 4 ✓

c. 6

d. 8

$$\frac{dy}{dt} = 3t^2 - 18t + 24 = 3(t - (2))(t - (4)) = 0 \text{ gives } t = 2 \text{ and } t = 4; \text{ the larger is } 4.$$

59 For $x(t) = t^2 - 2t$, $y(t) = t^3 + 9t^2 + 24t$, find the larger value of t where the tangent line is horizontal.

a. -2 ✓

b. -6

c. 8

d. -4

$$\frac{dy}{dt} = 3t^2 + 18t + 24 = 3(t - (-4))(t - (-2)) = 0 \text{ gives } t = -4 \text{ and } t = -2; \text{ the larger is } -2.$$

60 For $x(t) = t^2 - 3t$, $y(t) = t^3 + 3t^2 - 24t$, find the larger value of t where the tangent line is horizontal.

a. -4

b. -2

c. -8

d. 2 ✓

$$\frac{dy}{dt} = 3t^2 + 6t - 24 = 3(t - (-4))(t - (2)) = 0 \text{ gives } t = -4 \text{ and } t = 2; \text{ the larger is } 2.$$

61 For $x(t) = t^2 + 5t$, $y(t) = t^3 - 27t$, find the larger value of t where the tangent line is horizontal.

a. 3 ✓

b. 0

c. -9

d. -3

$\frac{dy}{dt} = 3t^2 - 27 = 3(t - (-3))(t - (3)) = 0$ gives $t = -3$ and $t = 3$; the larger is 3.

62 A particle moves with position $x(t) = t^2$, $y(t) = 7t$. Find its speed at $t = 12$.

a. 7

b. 25 ✓

c. 24

d. 27

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 7$. At $t = 12$: $\frac{dx}{dt} = 24$. Speed = $\sqrt{24^2 + 7^2} = \sqrt{625} = 25$.

63 A particle moves with position $x(t) = t^2$, $y(t) = 6t$. Find its speed at $t = 4$.

a. 12

b. 8

c. 6

d. 10 ✓

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6$. At $t = 4$: $\frac{dx}{dt} = 8$. Speed = $\sqrt{8^2 + 6^2} = \sqrt{100} = 10$.

64 A particle moves with position $x(t) = t^2$, $y(t) = 3t$. Find its speed at $t = 2$.

a. 3

b. 5 ✓

c. 7

d. 4

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3$. At $t = 2$: $\frac{dx}{dt} = 4$. Speed = $\sqrt{4^2 + 3^2} = \sqrt{25} = 5$.

65 A particle moves with position $x(t) = t^2$, $y(t) = 8t$. Find its speed at $t = 3$.

a. 6

b. 10 ✓

c. 8

d. 12

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 8$. At $t = 3$: $\frac{dx}{dt} = 6$. Speed $= \sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

66 A particle moves with position $x(t) = t^2$, $y(t) = 12t$. Find its speed at $t = 8$.

a. 22

b. 20 ✓

c. 16

d. 12

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 12$. At $t = 8$: $\frac{dx}{dt} = 16$. Speed $= \sqrt{16^2 + 12^2} = \sqrt{400} = 20$.